

name@gmail.com | portfolio.com | github.com/name

Skills

Programming Languages: Python, SQL

Deep Learning: Transformers, LLMs, PyTorch, OpenAI API, TensorFlow, Keras, Multiple GPU training, Parameter-efficient Fine-tuning, supervised fine-tuning

Machine Learning: scikit-learn, clustering, k-means, XGBoost

Artificial Intelligence: NumPy, Pandas, Natural language processing, Natural language generation, Prompt engineering, RAG, AI Agents, Computer vision, Text preprocessing

Databases: Microsoft SQL Server, PostgreSQL

Experience

AI Engineer,  Jan 2022 – Oct 2023

- Created text generation LLM by fine-tuning T5 on additional text-to-text tasks, defined a mitigation strategy as a new task and an anti semitic post as an input. The model generated texts to mitigate anti semitic posts on social media, the model produced text that is grammatically correct, logically structured, creative, and contextually accurate.
- Developed a state-of-the-art BERT-based classification model to improve the accuracy of a TF-IDF rule-based classifier, which had been in production since 2016. I fine-tuned BERT for text classification, resulting in 0.94 accuracy, improving the previous TF-IDF model by 0.14.
- Developed an efficient text preprocessing pipeline, conducted an extensive text cleaning, utilizing s-BERT for state-of-the-art text embedding, k-means clustering, and developed a creative algorithm based on pairwise cosine similarity, to detect a zero day anti semitic discourse weaves in a huge volume of data. Resulting in a list of unique and diverse representative sentences covering all the main topics in every cluster, detecting zero day anti semitic posts that the company's classifier failed to detect.

Research Assistant,  Jul 2021 – Jan 2022

- Designed hierarchical classification of images over CIFAR-100 dataset, utilizing state-of-the-art deep convolutional neural networks, designed to reflect the hierarchy, by stacking multiple CNN layers on 2 layers, the 1st layer corresponding to the 1st level of hierarchy, and the 2nd layer correspond to the 2nd level of the hierarchy. Resulting in improved classification results, and providing the network with human-like logic
- Architected and implemented efficient and persistent software solutions, resulting in easy optimization, usage, testing, reusing, and collaboration with the research doctor.

Computer Science Tutor,  Jul 2019 – Jan 2022

- Developed strong problem solving skills and techniques to tackle problems, by solving hundreds of leetcode challenges, And demonstrated to students how to translate them to their own problems, providing them with tools to handle future problems, results in high demand for private lessons, and sessions before exams for a few years after.

Projects

Deep Freeze  name.com/projectdemo

- Developed offensive ransomware to encrypt the target personal files with AES-128 asymmetric key, leveraging RSA-2048 to encrypt the target key or to decrypt his key after bitcoin payment, resulting in unbreakable ransom

Face Mask Object Detection  github.com/name/project

- Developed an object detection to track not wearing face mask on COVID-19 period, utilizing light weight convolutional deep neural network to implement the solution on cheap serverless Raspberry Pi, with heat sensors to track potential COVID spreaders, leveraging "Viola-Jones" for extra confidence, resulting in 0.95 accuracy on real-world predictions.

Education

 BS in Computer Science, GPA: 95

Aug 2021